Yining Wu

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Research Interests: (1) **LLMs & AI Agents:** Developing adaptive agents and improving model reasoning, learning, and decision-making through feedback, post-training, and multi-turn interactions. (2) **Affective Computing:** Modeling and controlling emotional behaviors in LLM-based agents to support realistic, personalized human-AI interactions. (3) **AI for Healthcare:** Studying how LLMs interpret clinical evidence, support medical decision-making, and enable realistic healthcare simulations.

EDUCATION

University of Texas at Austin Aug 2025 – May 2027
M.S. in Information Studies; GPA: 4.0/4.0 *Austin, TX*
Coursework: Machine Learning, Deep Learning, Natural Language Processing, Post-Training of LLMs, AI in Health

Tongji University Sept 2020 – Jun 2025
B.Eng. in Vehicle Engineering; GPA: 3.97/4.0 *Shanghai, China*

PUBLICATIONS & MANUSCRIPTS

Peer-Reviewed Conference Papers

[P1] **Wu, Y.**, Du, T., Fang, J., Zhang, C., Admane, S., Ding, Y. (2026). EmoPatient: An Emotion-Directed Patient Simulator for Realistic Palliative Care Communication Training. *2026 American Medical Informatics Association (AMIA) Annual Symposium*, Dallas, TX, USA. (accepted main conference paper).

Preprints & Work Under Review

[M1] **Wu, Y.**, Li, P., Fang, J., Tao, C., Ding, Y. (2027). Same Diagnosis, Different Evidence Maps: A "Rashomon Effect" in Clinical Evidence Appraisal. *2027 IEEE International Conference on Healthcare Informatics (IEEE ICHI)*, Washington, DC, USA. (manuscript submitted).

[M2] **Wu, Y.**, Hosseinpour, R., Bareither, M., Du, T., Pratt, C., Ding, Y. (2027). Evaluating the Efficacy of Emotion Steering in LLM-Based Virtual Patients. *2027 IEEE International Conference on Healthcare Informatics (IEEE ICHI)*, Washington, DC, USA. (manuscript submitted).

[M3] Xu, J., Xiong, Y., **Wu, Y.**, Fang, J., Kang, E., Han, Y., Wang, P., Li, P., Ding, Y. (2026). DxSkill: How Feedback Shapes Clinical Agents' Learning from Experience. *ACL Rolling Review (ARR)*, October 2026. (manuscript submitted).

[M4] Fang, J., Chen, R., **Wu, Y.**, Xu, J., Yang, X., Yu, J., Shi, W., Ding, Y., Zhang, Y. (2026). ROUNDS: Source-Framed Diagnostic Revision after Multi-Turn Clinical Evidence. *2026 Machine Learning for Health (ML4H) Symposium*, Sydney, Australia. (manuscript submitted).

Workshop Papers & Abstracts

[W1] **Wu, Y.**, DiGiacomo, P., Ding, Y., Brode, W. (2026). Grounded in Consensus, In Step With Emerging Science: A Consensus-Anchored Multi-Corpus Clinical Chatbot for Long COVID. *Workshop on LLM/VLM Deployment Opportunities and Risks in Healthcare (DAIH), at the 2026 Conference on Language Modeling (COLM)*, San Francisco, CA, USA. (accepted workshop paper).

[W2] Yang, X., Sha, Z., Li, J., Yu, J., Sun, Y., Zhao, M., Fang, J., Guo, X., **Wu, Y.**, Hu, X., Luo, Y., Liu, Q., Wang, Z. (2026). No Hidden Prompts Needed! You Can Game AI Peer Review with Presentation-Only Revisions. *AI-Native Academia: Authorship, Peer Review, and Conference Governance under AI Workshop, 40th Annual Conference on Neural Information Processing Systems (NeurIPS 2026)*, Atlanta, GA, USA. (accepted workshop paper).

[W3] Brode, W. M., **Wu, Y.**, DiGiacomo, P., Quilantan, J., Jr., Raghav, A., Verduzco-Gutierrez, M., Li, J. J., Saiyed, S., Ding, Y. (2027). Keeping Pace with an Uncertain Evidence Base: Living, Traceable Evidence Infrastructure for Long COVID Clinical Decision Support. *Translational Science 2027 (ACTS)*, Washington, DC, USA, April 2027. (abstract submitted).

RESEARCH EXPERIENCE

DxSkill: How Feedback Shapes Clinical Agents' Learning from Experience [M3] Sep 2026 – Present
Research Assistant; AI Health Lab, School of Computing, UT Austin; Supervisor: Prof. Ying Ding *Austin, TX*

- Curated source-traceable excerpts from WSES Jerusalem, Tokyo, DGVS/DGAV, and ACG clinical guidelines to provide structured feedback for clinical agents' post-encounter reflection and diagnostic skill revision.
- Developing stage-wise skill-fusion methods to consolidate complementary skills, resolve overlap across learning stages, and maintain coherent, reusable diagnostic skill libraries.

Emotion-Aware and Controllable LLM Virtual Patients Jan 2026 – Oct 2026
Research Assistant; AI Health Lab, School of Computing, UT Austin; Supervisor: Prof. Ying Ding *Austin, TX*

Phase I: Emotion-Adaptive Patient Simulation for Palliative Care [P1]

- Compiled a cohort of 354 de-identified patients with advanced metastatic malignancies from MIMIC-IV and MIMIC-IV-Note, and developed clinically grounded patient personas using diagnosis-code filtering and SQL extraction.
- Designed a closed-loop multi-agent architecture for turn-level emotion estimation and response control; led evaluations across

20 clinical cases, 60 personality-conditioned profiles, and four LLM backbones, including robustness analyses and ablations.

Phase II: Steerable Emotional Behavior in LLM Virtual Patients [M2]

- Developed an emotion-steering framework for LLM-based virtual patients, implementing prompt-based strategies and contrastive activation-vector injection to control fear, sadness, and anger at different intensities in multi-turn clinical dialogues.
- Led large-scale evaluations of emotion controllability and clinical fidelity across 927 held-out clinician turns, identifying trade-offs between target-emotion alignment, intensity control, and preservation of patient-specific clinical information.

Evaluating Clinical Evidence Appraisal across LLMs with Differential-Evidence Maps [M1] Jul 2026 – Oct 2026
Jointly advised by Prof. Ying Ding at UT & Prof. Cui Tao at Mayo Clinic *Austin, TX / Jacksonville, FL*

- Constructed SCT-inspired differential-evidence maps, a three-layer concordance framework, and a six-role evidence taxonomy to evaluate clinical evidence appraisal across eight LLMs and 322 MedQA/MedMCQA cases.
- Identified substantial evidence-level disagreement across 28 model pairs despite shared correct diagnoses, and validated the diagnostic relevance of evidence ratings through self-ablation and bootstrap analysis.

Consensus-Anchored Clinical Decision Support for Long COVID [W1] Oct 2025 – Jun 2026
Graduate Research Assistant; Dell Medical School, UT Austin; Supervisor: Prof. William Micheal Brode *Austin, TX*

- Engineered a multi-corpus clinical chatbot integrating expert guidance, PubMed literature, ClinicalTrials.gov records, and systematic reviews through specialized retrieval, evidence synthesis, and citation-tracing pipelines.
- Implemented consensus-anchored response generation and coordinated clinician-in-the-loop evaluations of clinical reliability, utility, completeness, and safety.

PROFESSIONAL EXPERIENCE

Mayo Clinic May 2026 – Aug 2026
Data Science & AI Research Intern; Department of Artificial Intelligence & Informatics *Jacksonville, FL*

Project: Post-Training LLMs for Post-Transplant Cardiovascular Risk Prediction

- Built an LLM pipeline predicting five-year MACE from EHR histories of 4,534 liver transplant recipients.
- Built an extractor–scorer for multimodal timelines; post-trained Qwen2.5 with LoRA-SFT, GRPO, and OPD.
- Benchmarked five pipeline variants against conventional ML and deep-learning baselines for MACE prediction.

Project: TrialAgent: A Multi-Agent Framework for Automating EHR-Based Clinical Trial Emulation

- Built a benchmark and agent evaluation harness for translating clinical trial protocols into computable EHR study definitions, including eligibility criteria, cohort construction, and endpoint specification.
- Designed annotation and adjudication workflows for consistent protocol-to-cohort evaluation across agentic systems.
- Contributed to the end-to-end agent framework through figures, technical documentation, and manuscript development.

Volkswagen Group Jan 2025 – Jun 2025
Product Strategy Intern

- Translated user and market data into product-strategy recommendations for emerging automotive needs.
- Built a RAG persona pipeline combining semantic retrieval and LLM generation to consolidate fragmented user evidence.
- Automated persona consistency and redundancy metrics to enable iterative, data-driven pipeline refinement.

SKILLS

Programming, ML & Data: Python, SQL, PyTorch, Transformers, scikit-learn, pandas, NumPy, Git, Linux
Foundation Models & Post-Training: SFT, LoRA, preference optimization, reinforcement learning, knowledge distillation
Agentic AI: RAG, multi-agent systems, tool use, harness engineering, error analysis, human-in-the-loop evaluation

SELECTED PRESENTATIONS

[1] *EmoPatient: An Emotion-Directed Patient Simulator for Realistic Palliative Care Communication Training*.
- Oral presentation, AMIA 2026 Annual Symposium.
- Poster, UT System AI in Healthcare Symposium.
- Presentation, 2026 Workshop on Interactive Systems in Healthcare (WISH) at AMIA.

[2] *Grounded in Consensus, In Step With Emerging Science: A Consensus-Anchored Multi-Corpus Clinical Chatbot for Long COVID*. Poster, DAIH Workshop at COLM 2026.

HONORS & AWARDS

Professional Development Award ($1,000), UT Austin School of Computing, College of Natural Sciences 2026
DAIH Travel Award, DAIH Workshop at COLM 2026
Outstanding Undergraduate Scholarship, Tongji University 2020 – 2024

ACADEMIC SERVICE

Reviewer. GenAI4Health Workshop at NeurIPS 2026